

Your First Model

EES 4760/5760

Agent-Based and Individual-Based Computational Modeling

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Homework

Homework:

- In the mushroom hunt, were there always 80 red patches?
- Any questions about modified mushroom hunt model?
- Let's talk about ODD exercise.

Writing a model from an ODD

- Questions about writing a model from Butterfly ODD?
- Were there things the ODD was unclear about?

Butterfly Model

Download the model

- Download butterfly model from ees4760.jmgilligan.org/models/class_05/butterfly_model_class_5.nlogox
 - Or go to the “Downloads” page on the class web site ees4760.jmgilligan.org, click on “5. Butterfly Models” and download the “Basic Butterfly Model”
 - You can download the final enhanced butterfly model we will write in class from “Enhanced Butterfly Model”, or https://ees4760.jmgilligan.org/models/class_05/butterfly_model_class_5.nlogox

Enhancing the Butterfly model

- Put a slider for q
 - In the “Code” page:
 - Remove q from the global variables
 - Remove the initialization of q from `to setup`
- Add patches-own variable to indicate whether it was visited.

```
patches-own [  
  elevation  
  visited? ; question mark reminds us it's a true/false variable  
]  
  
to setup  
  ...  
  ask patches [  
    set visited? false  
  ]  
  ...  
end
```

- Add turtles-own variable to remember the patch where it started.

```
turtles-own [  
  origin  
]
```

Enhancing the Butterfly model

- Put a slider for q
- Add patches-own variable to indicate whether it was visited.
- Add turtles-own variable to remember the patch where it started.
- Set the number of butterflies to 50.
- Stop butterfly from moving if it's at the top of a hill.
 - How can you tell whether it's on the top?

Enhancing the Butterfly model

- Write a reporter for corridor width

$$\text{Corridor width} = \frac{\# \text{ patches visited}}{\text{distance from start}}$$

- Put an **observer** on the interface
- Define a reporter:

```
to-report corridor-width  
  let pcount count patches with [visited?]  
  let dist mean [distance origin] of turtles  
  report pcount / dist  
end
```

- Is there a problem when you hit “Setup”?

Running Experiments: BehaviorSpace

- Vary any parameter that has a control on the model's interface
- Writes output to `.csv` spreadsheet file
 - Four options:
 1. Spreadsheet
 2. Table
 3. Statistics
 4. Lists (only useful if you also choose spreadsheet or table)
- Note: Data written in spreadsheet or table might be out of order.

Behaviorspace Table Format

	A	B	C	D
1	BehaviorSpace results (NetLogo 6.4.0) Table version 2.0			
2	enhanced_butterfly_model_class_5.nlogo			
3	vary-q-all-steps			
4	09/03/2024 21:47:41:586 -0500			
5	min-pxcor	max-pxcor	min-pycor	max-pycor
6	0	149	0	149
7	[run number]	q	[step]	corridor-width
8	3	0	0	35.35533906
9	3	0	1	41.01515439
10	15	0	0	35.35533906
11	13	0	0	35.35533906
12	10	0	0	35.35533906
13	14	0	0	35.35533906
14	3	0	2	49.51907135
15	2	0	0	35.35533906

Behaviorspace Spreadsheet Format

	A	B	C	D	E	F	G	H	I
1	BehaviorSpace results (NetLogo 6.4.0) Spreadsheet version 2.0								
2	enhanced_butterfly_model_class_5.nlogo								
3	vary-q-all-steps								
4	09/03/2024 21:48:10:612 -0500								
5	min-pxcor	max-pxcor	min-pycor	max-pycor					
6	0	149	0	149					
7	[run number]	1	1	2	2	3	3	4	4
8	q	0		0		0		0	
9	[reporter]	[step]	corridor-width	[step]	corridor-width	[step]	corridor-width	[step]	corridor-width
10	[final]	999	423.4725	999	438.6560	999	421.6216	999	418.1058
11	[min]	0	35.3553	0	35.3553	0	35.3553	0	35.3553
12	[max]	999	424.0267	999	438.9208	999	421.8210	999	418.3716
13	[mean]	499.5	300.8170	499.5	303.9720	499.5	302.6952	499.5	300.0260
14	[total steps]	999	999	999	999	999	999	999	999
15									
16	[all run data]	[step]	corridor-width	[step]	corridor-width	[step]	corridor-width	[step]	corridor-width
17		0	35.3553	0	35.3553	0	35.3553	0	35.3553
18		1	40.0116	1	40.6208	1	41.0152	1	39.9840
19		2	47.1075	2	47.1339	2	49.5191	2	50.3201
20		3	49.0780	3	49.6450	3	48.9961	3	52.1140
21		4	50.7408	4	51.5005	4	49.7353	4	55.5346
22		5	49.0910	5	52.4445	5	50.0928	5	55.9192
23		6	51.0026	6	54.4283	6	54.3610	6	59.1715
24		7	51.2436	7	59.8473	7	58.3433	7	61.3543
25		8	54.9979	8	59.8388	8	60.3152	8	62.3758
26		9	58.8774	9	60.6895	9	61.9538	9	64.0257
27		10	60.4508	10	60.4804	10	63.6785	10	63.8910

Behaviorspace Stats Format

	A	B	C	D
1	BehaviorSpace results (NetLogo 6.4.0)		Stats version 2.0	
2	enhanced_butterfly_model_class_5.nlogo			
3	vary-q-all-steps			
4	09/03/2024 21:48:11:511 -0500			
5	min-pxcor	max-pxcor	min-pycor	max-pycor
6	0	149	0	149
7	q	[step]	(mean) corridor-width	(std) corridor-width
8	0	0	35.3553	0.0000
9	0	1	40.3559	0.4474
10	0	2	47.8396	1.3203
11	0	3	48.7062	1.7825
12	0	4	51.2488	2.2297
13	0	5	53.2222	2.3226
14	0	6	56.0615	2.3681
15	0	7	58.0982	2.2659

Analyzing Behaviorspace Output

- Behaviorspace output format is annoying.
 - Each line is some tick of some run.
 - How to organize, and average over runs?
 - The new *stats* output helps with this, though.
- analyzeBehaviorspace app:
 - On the web at <https://ees4760.jmgilligan.org/analyze-behaviorspace>
 - Install on your own computer using R.
 - Instructions at <https://github.com/j-magnolia/analyzeBehaviorspace>.
 - After installing:

```
library(analyzeBehaviorspace)
launch_abs()
```

AnalyzeBehaviorspace

Analyze BehaviorSpace Experiments

Choose CSV File

Browse...

No file selected

x axis

y axis

Group by

☒ points

☐ lines

Std. Dev.

☒ none ☐ error bars ☐ bands

☐ Only show last tick

☐ Summary table?

⬇ Save Plot (session/05fab11f6c9725f7eb5252a5f8dd850a/download/save_plot?w=)

⬇ Save Table (session/05fab11f6c9725f7eb5252a5f8dd850a/download/save_table?w=)

Quit

Rename variables

from

Emergence

Emergence

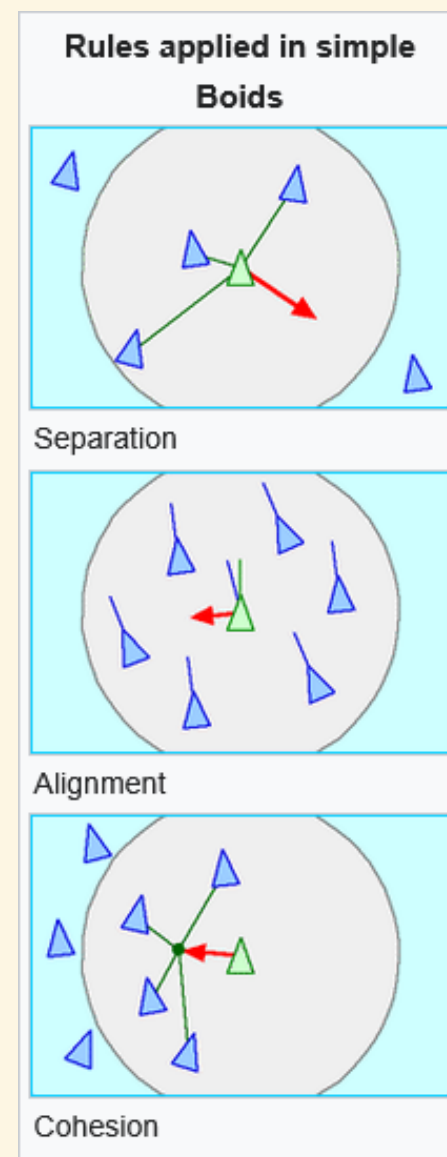
- A tricky concept.
- Joshua Epstein in *Growing Artificial Societies*: “stable macroscopic patterns arising from the local interaction of agents.”
- Epstein ten years later: “I have always been uncomfortable with the vagueness and occasional mysticism surrounding this word.”
- Epstein now prefers to talk about “*Generative Social Science*” instead of “*emergence*”

Examples of Emergence

Simple Flocking

Simple Flocking

- How do birds coordinate their flight to move in coherent flocks?
- Craig Reynolds: [boids](#) 1989
 - boids only see nearby boids within some radius
 - Three principles of motion:
 1. Separation
 - Turn away if too close to another boid
 2. Alignment
 - Turn toward the mean direction neighbors are flying
 3. Coherence
 - Turn toward mean position of neighbors (center)



Computer Graphics, Volume 21, Number 4, July 1987

Flocks, Herds, and Schools: A Distributed Behavioral Model

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Abstract

The aggregate motion of a flock of birds, a herd of land animals, or a school of fish is a beautiful and familiar part of the natural world. But this type of complex motion is rarely seen in computer animation. This paper explores an approach based on simulation as an alternative to scripting the paths of each bird individually. The simulated flock is an elaboration of a particle system, with the simulated birds being the particles. The aggregate motion of the simulated flock is created by a distributed behavioral model much like that at work in a natural flock; the birds choose their own course. Each simulated bird is implemented as an independent actor that navigates according to its local perception of the dynamic environment, the laws of simulated physics that rule its motion, and a set of behaviors programmed into it by the "animator." The aggregate motion of the simulated flock is the result of the dense interaction of the relatively simple behaviors of the individual simulated birds.

it seems randomly arrayed and yet is magnificently synchronized. Perhaps most puzzling is the strong impression of intentional, centralized control. Yet all evidence indicates that flock motion must be merely the aggregate result of the actions of individual animals, each acting solely on the basis of its own local perception of the world.

One area of interest within computer animation is the description and control of all types of motion. Computer animators seek both to invent wholly new types of abstract motion and to duplicate (or make variations on) the motions found in the real world. At first glance, producing an animated, computer graphic portrayal of a flock of birds presents significant difficulties. Scripting the path of a large number of individual objects using traditional computer animation techniques would be tedious. Given the complex paths that birds follow, it is doubtful this specification could be made without error. Even if a reasonable number of suitable paths could be described, it

C.W. Reynolds, "Flocks, Herds, and Schools", SIGGRAPH'87 25-34 (1987) doi: 10.1145/37401.37406

Boids in NetLogo

 powered by NetLogo

flocking

File:

New

Revert to Original

Export:

NetLogo

HTML

 Mode: Interactive

Commands and Code: Bottom

-

Model Speed

+

?

 for shortcuts

ticks:

population

300

setup

go 

vision

5

patches

minimum-separation

1

patches

max-align-turn

5

degrees

max-cohere-turn

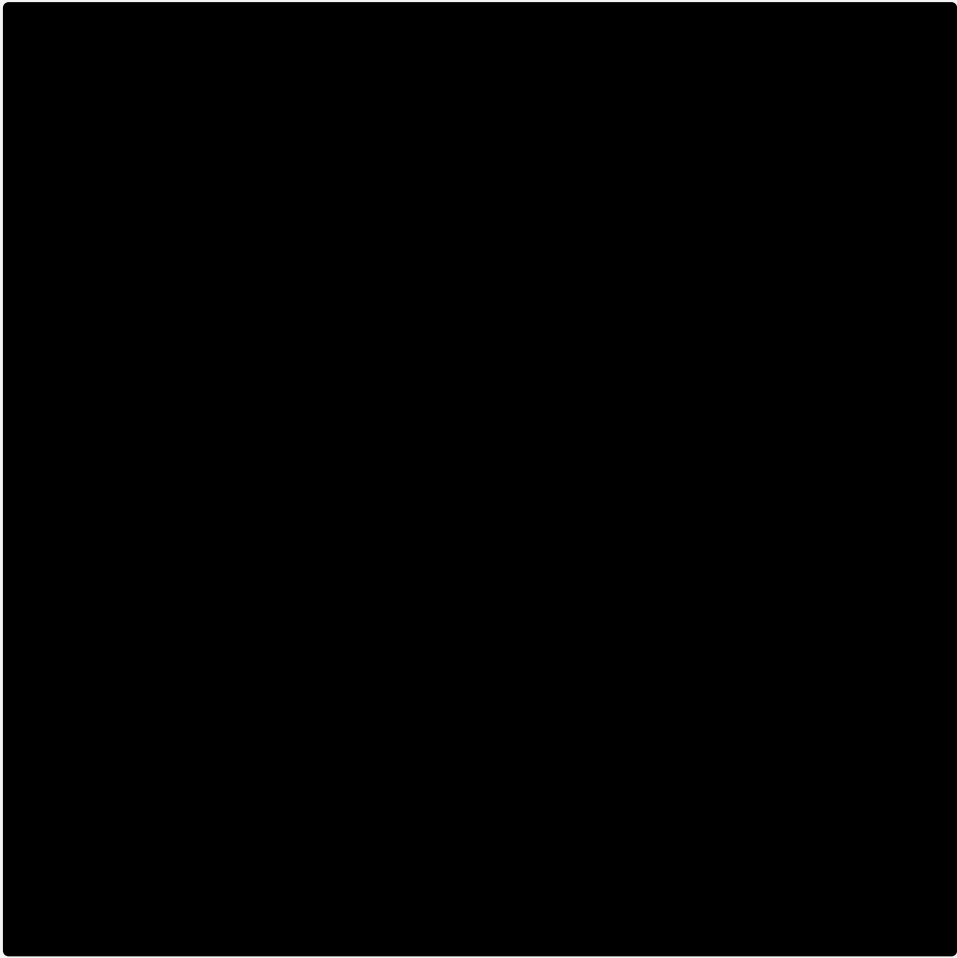
3

degrees

max-separate-turn

1.5

degrees



Command Center

▼

NetLogo Code

▼

Model Info

▼

Agent Inspection

▼

Geese Flocking

 powered by NetLogo

Flocking Vee Formations

File: New Revert to Original
Export: NetLogo HTML

 Mode: Interactive

Commands and Code: Bottom

-

Model Speed

+

?

 for shortcuts

ticks:

number-of-birds

40

setup

go

Vision Parameters

vision-distance

10

vision-cone

120

deg

obstruction-co...

45

deg

Motion Parameters

base-speed

0.2

speed-change-f...

0.15

updraft-distance

4

too-close

1.6

max-turn

4

deg

Visualization

show-unhappy?

Command Center

NetLogo Code

Complex Flocking: Starlings

Starling Murmuration

- Thousands of individuals
 - unique and different
 - interact locally
 - show adaptive behavior

Behavioral Ecology
doi:10.1093/beheco/arq149

Self-organized aerial displays of thousands of starlings: a model

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Through combining theoretical models and empirical data, complexity science has increased our understanding of social behavior of animals, in particular of social insects, primates, and fish. What are missing are studies of collective behavior of huge swarms of birds. Recently detailed empirical data have been collected of the swarming maneuvers of large flocks of thousands of starlings (*Sturnus vulgaris*) at their communal sleeping site (roost). Their flocking maneuvers are of dazzling

Starling murmuration



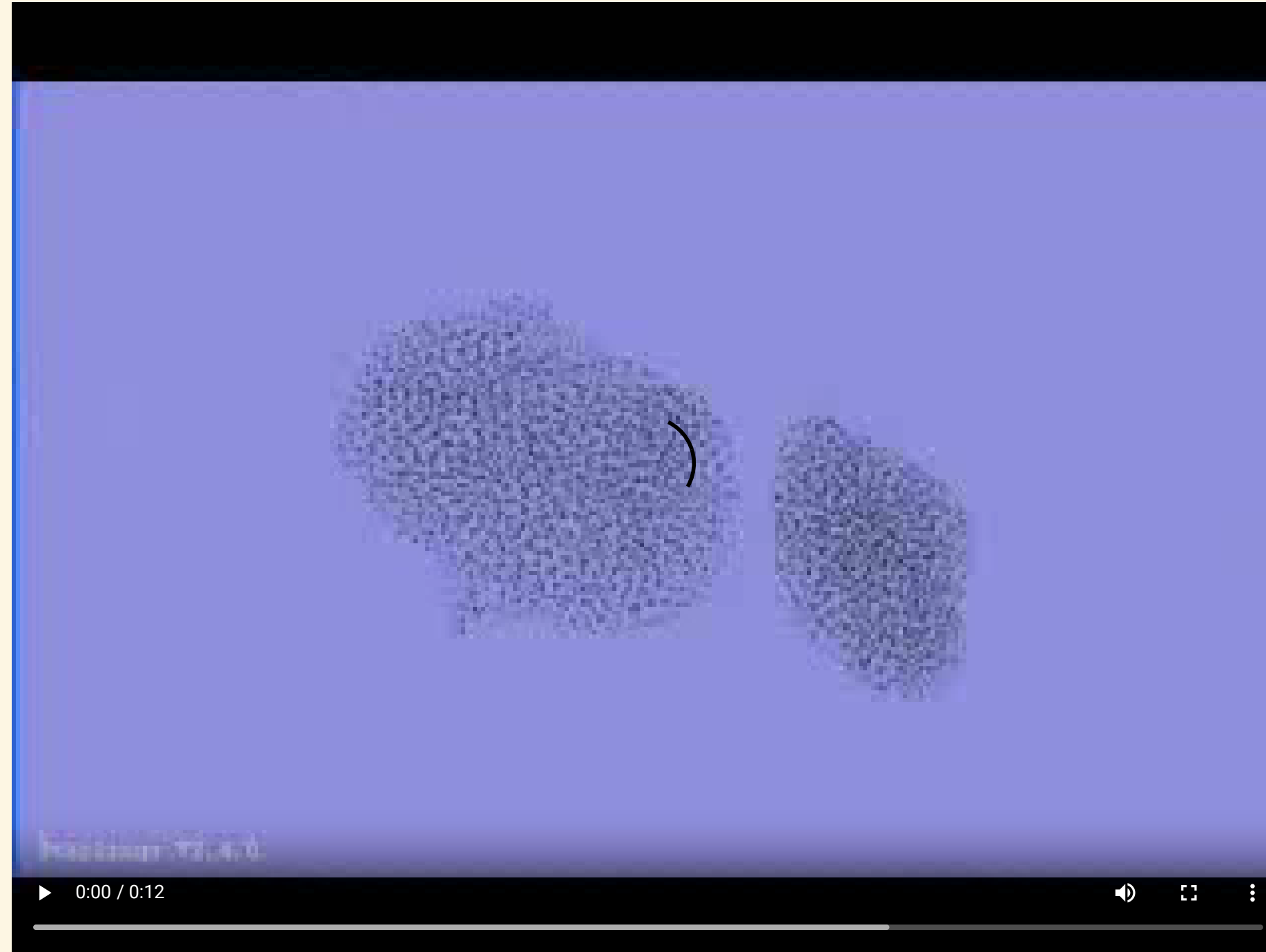
By Liberty Smith & Sophie Windsor Clive, Islands and Rivers, <https://vimeo.com/31158841>

Flock of thousands of starlings



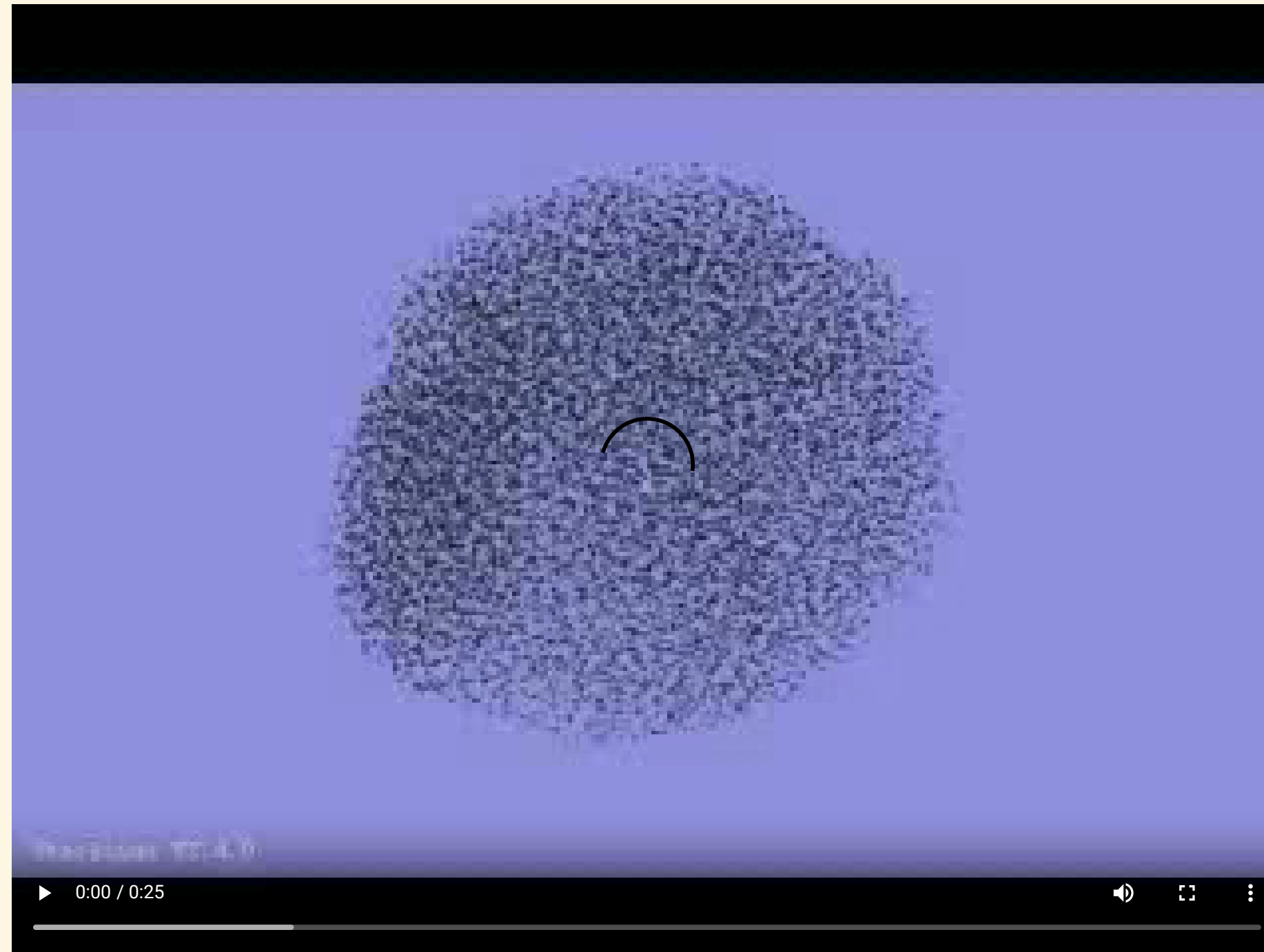
H. Hildenbrandt, C. Carere, & C.K. Hemelrijk, *Behavioral Ecology*, **21**, 1349. DOI: 10.1093/beheco/arq149

Simulated flock of thousands of starlings



H. Hildenbrandt, C. Carere, & C.K. Hemelrijk, *Behavioral Ecology*, **21**, 1349. DOI: 10.1093/beheco/arq149

Simulated flock of thousands of starlings



H. Hildenbrandt, C. Carere, & C.K. Hemelrijk, *Behavioral Ecology*, **21**, 1349. DOI: 10.1093/beheco/arq149